

MAY 12-18, 2026 // 92 ENGINEERS // 3 CONTINENTS

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01  
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# The AI Productivity *Paradox.*

Why teams shipping 30× more code with AI still feel slower — and what **92 Spring Boot engineers** say is actually holding them back in 2026.

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92

COMPLETED RESPONSES

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43%

SURVEY CONVERSION

---

  
21.9%

BIGGEST FRUSTRATION:  
SLOW BUILD & TEST  
FEEDBACK

---

  
25%

BLAME AI-GENERATED  
CODE ITSELF

## § 01 / EXECUTIVE SUMMARY

# The promise doesn't match the pipeline.

AI tools promised to make us 10× faster. The largest independent studies of 2025 and 2026 tell a more uncomfortable story — and our own data from 92 engineers confirms it.

## The two studies framing this report

Google's **2025 DORA Report** — surveying nearly 5,000 technology professionals — found that **90% of teams now use AI** and **80% feel more productive**, yet AI adoption still has a *negative* relationship with software delivery stability. The report's central conclusion: AI doesn't fix a team; it amplifies whatever is already there.

The **METR randomized controlled trial** (2025) measured experienced open-source developers on real tasks. Result: they thought AI made them **20% faster**. It actually made them **19% slower**. The bottleneck was reviewing, debugging, and verifying generated code — precisely the work that gets harder on a flaky test pipeline.

## What our 92 engineers say

The same paradox shows up in our data. **30%** name quality and reliability as their absolute highest priority — ahead of speed. But the things blocking them are not product complexity. They are **slow tests, flaky pipelines, unreviewed AI output, and legacy code drowning in technical debt**.

For Spring Boot teams, the picture sharpens: **19%** say their biggest testing bottleneck is simply *getting everyone on the same testing strategy*. Tools are not the constraint. **Alignment, integration tests, and feedback loops are.**

## Four numbers to remember

**21.9%**

SAY LONG BUILD &  
TEST TIMES ARE  
THEIR #1  
FRUSTRATION

**25%**

FRUSTRATED BY  
ANALYZING &  
DEBUGGING  
AI-GENERATED  
CODE

**21.4%**

BLOCKED BY  
UNMANAGEABLE  
LEGACY CODE

**19.0%**

SPRING BOOT  
TEAMS: NO  
SHARED TESTING  
STRATEGY

“ We feel like we are generating 30× more code using AI. But how do we reliably secure this code without our test pipelines turning into a maintenance time bomb?

— THE QUESTION BEHIND THIS STUDY

## § 02 / METHODOLOGY

# How the numbers were gathered.

Six questions. Three minutes. One week of fieldwork across a self-selected audience of working Java and Spring Boot engineers.

## Fieldwork

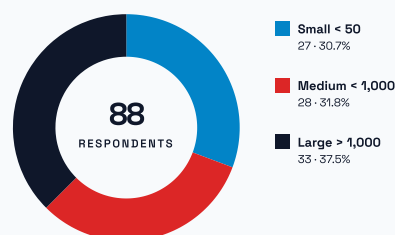
| FIELD               | VALUE                |
|---------------------|----------------------|
| Period              | 12 - 18 May 2026     |
| Survey visits       | 214                  |
| Completed responses | 92                   |
| Completion rate     | 43.0%                |
| Leads captured      | 60                   |
| Channels            | Newsletter, LinkedIn |
| Questions           | 6 (4 open · 2 multi) |

## A note on the sample

The audience self-selected via a developer newsletter and LinkedIn distribution focused on Java and Spring Boot. The sample skews toward practitioners actively thinking about testing and AI — not the “not-using-AI-yet” long tail. Read the findings as a snapshot of *engaged, AI-aware* engineering teams rather than the global average.

### Company size of respondents

N = 88 · QUESTION 5



### WHY THIS MATTERS

The respondent base is split almost evenly across small, mid-market and enterprise. Findings hold whether you ship a 5-developer SaaS or operate a Spring Boot estate inside a Fortune 500.

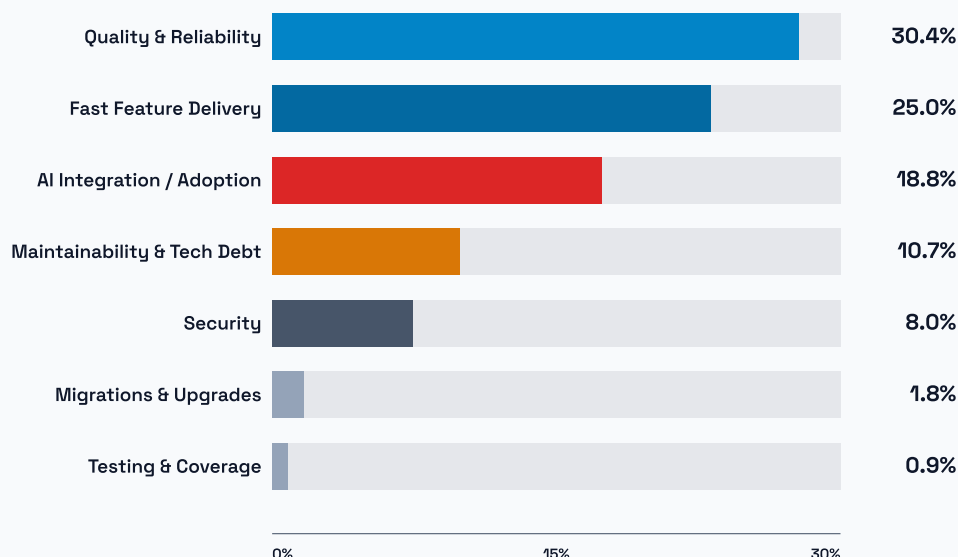
## § 03 / WHAT'S TOP OF MIND

# Quality still wins.

We asked 112 engineers what their team's absolute highest priority is right now. Free-text answers, hand-coded into seven themes. *Some answers map to more than one.*

## Q1 — “What is currently the absolute highest priority for you and your engineering team?”

N = 112 FREE-TEXT ANSWERS · MULTI-CODED · SHARE OF RESPONDENTS MENTIONING THEME



“ Shipping features reliably with minimal or no technical debt, supported by tests that effectively identify regressions and expose bugs.

— RESPONDENT #81

## The signal

**Quality and speed are not opposites in this dataset.** Engineers want both — in the same answer. Almost a quarter of respondents explicitly paired them: “deliver fast *with* high quality”, “faster delivery with higher quality”, “ship features quicker while tackling technical debt.”

AI is named directly by **18.8%** — but always with the same hedge: “leverage AI, but in a controlled manner”; “use AI tools but review code”; “not reduce code quality while using Gen AI as much as possible”.

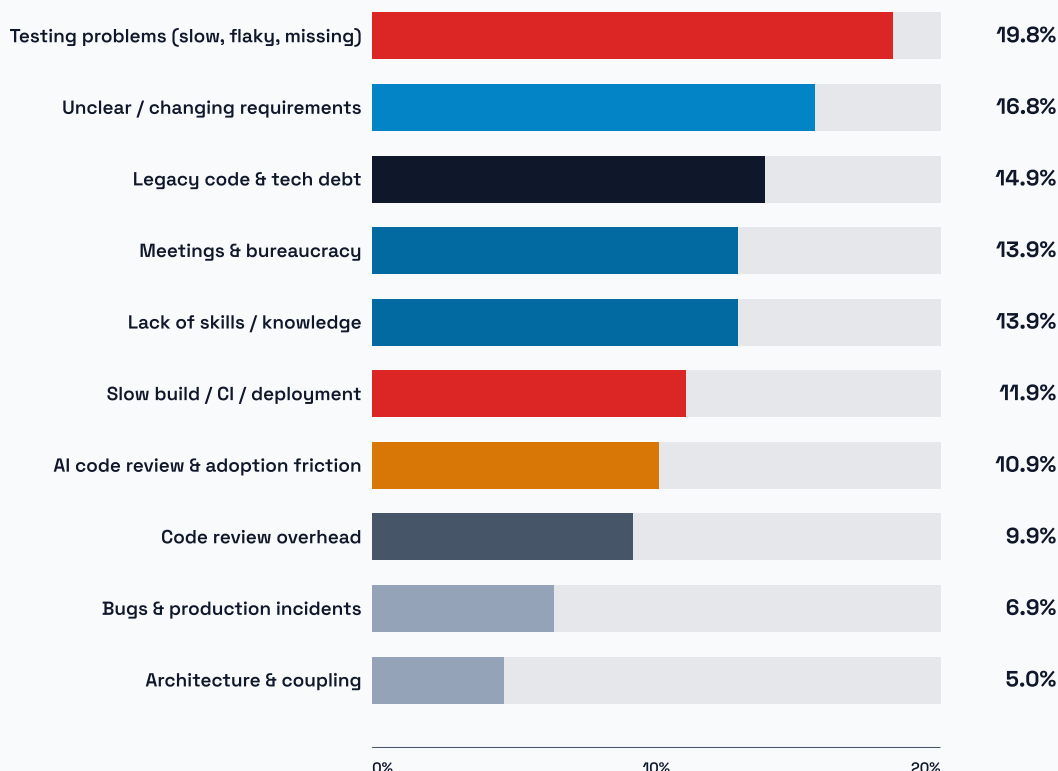
## § 04 / THE REAL BOTTLENECKS

# What is actually blocking us.

101 free-text answers to “what is holding you back from shipping features quickly?”, coded into ten themes. *One in five teams blames their test suite.*

## Q2 — What is currently holding you back the most from shipping new features quickly and smoothly?

N = 101 FREE-TEXT ANSWERS · MULTI-CODED · SHARE OF RESPONDENTS MENTIONING THEME



“ Slow and painful integration tests, dead code and unused functionalities, lack of observability.

RESPONDENT · #53

“ Review time gets longer due to bigger and more complicated AI-generated merge requests.

RESPONDENT · #12

“ Interpreting the tests to grasp dependencies is challenging — sometimes they’re just AI-generated noise.

RESPONDENT · #73

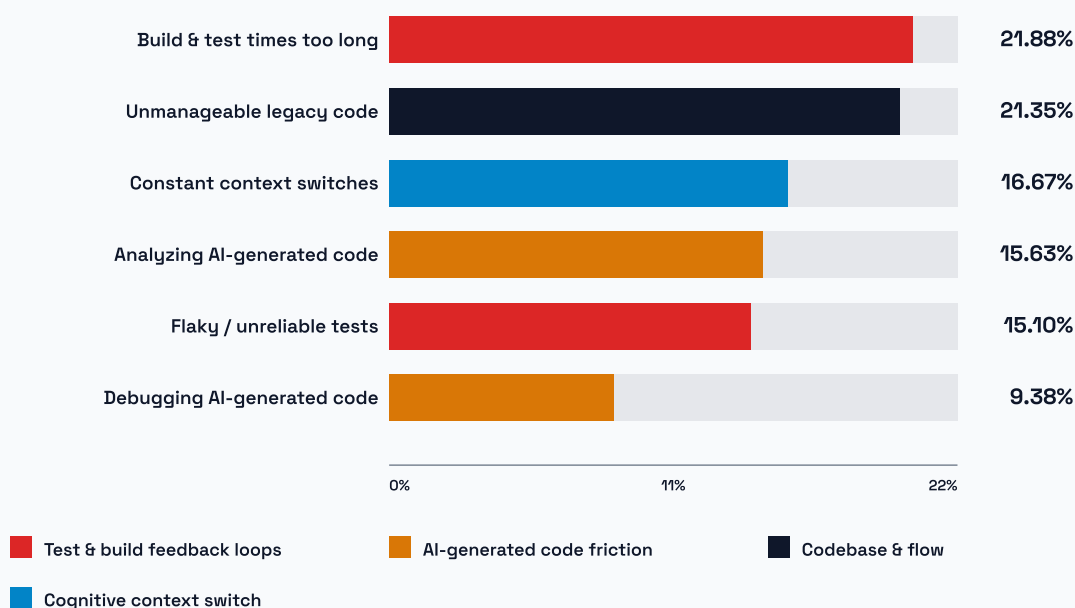
## § 05 / THE FRICTION MAP

# Where the day goes.

Forced-choice question: pick what frustrates you most about your current deployment and testing process. Multiple selections allowed.

## Q3 — What frustrates you the most about your current deployment or testing process?

192 SELECTIONS FROM 92 RESPONDENTS · % OF TOTAL SELECTIONS SHOWN



### THE HIDDEN MATH OF THIS CHART

Grouped by theme, the picture changes. **Test & build pipeline friction** (slow times + flaky tests) accounts for **36.98%** of all selections. **AI-generated code friction** (analyzing + debugging) accounts for **25.01%**. The two together — a productivity tax on every AI-assisted change — eat **three out of five** frustration votes in the sample.

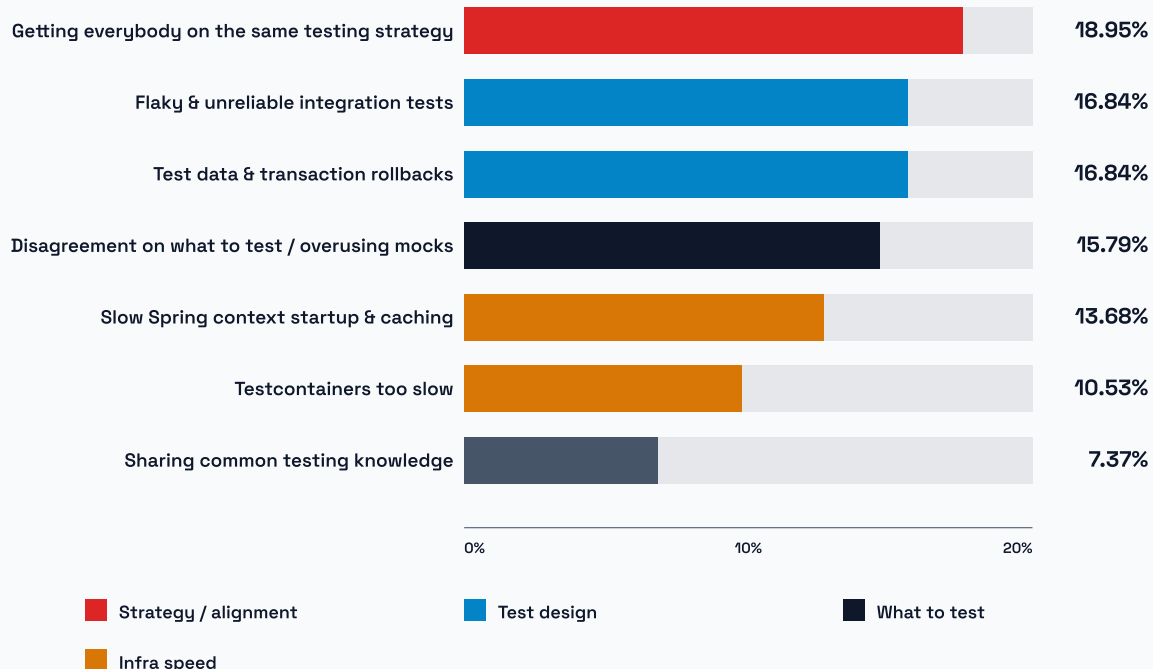
## § 06 / THE SPRING BOOT TESTING STACK

# Where Spring Boot teams get stuck.

For automated testing inside Spring Boot codebases, what is your team's biggest bottleneck right now? *Pick your top frustration.*

## Q6 — Spring Boot automated testing bottlenecks

N = 95 SELECTIONS · % OF TOTAL SELECTIONS SHOWN



## The pattern hiding in the data

The top three answers — strategy alignment (18.95%), flaky integration tests (16.84%), and test data management (16.84%) — are **not tooling problems**. Spring Boot, JUnit 5 and Testcontainers all *can* solve them.

What's missing is a shared **opinion** on how the team writes tests: when to use a slice test versus a full `@SpringBootTest`, when a Testcontainer is warranted, and how to keep state clean between tests.



### READ TOGETHER WITH Q3

22% of all engineers (Q3) say build & test times are their #1 frustration. For Spring Boot teams (Q6), **only 24%** blame infrastructure (Spring context + Testcontainers). The other **76%** blame strategy, design and discipline.

**You can't buy your way out of this with faster hardware.**

## § 07 / ANALYSIS

# The acceleration whiplash.

AI generates more code — faster. Everything *downstream* of that code generation has to absorb the new volume. Three independent datasets now point to the same place.

## 90%

ENGINEERS USING AI AT WORK (DORA 2025)

"AI adoption is near-universal". 80% report higher productivity, but 30% have little or no trust in AI-generated code.

## -19%

PRODUCTIVITY CHANGE FOR EXPERIENCED DEVS USING AI (METR RCT, 2025)

Developers *believed* AI made them 20% faster. The randomized controlled trial showed they were 19% slower on real tasks.

## +441%

MEDIAN TIME-IN-PR-REVIEW CHANGE AMONG 22K AI-USING DEVS (FAROS AI, 2026)

The throughput moved up; the review tax exploded behind it. Incidents-per-PR also rose 242%.

## What our 92 engineers are telling us

The DORA report calls AI a "*mirror and a multiplier*". Our data shows the mirror clearly. Engineers don't blame AI in the abstract — they blame the work AI creates downstream:

- **25% in Q3** point to AI-generated code itself — analyzing it (15.6%) and debugging it (9.4%).
- **11% in Q2** independently surface "AI conversations", "reviewing generated code", and "bigger, more complicated AI MRs" as a top blocker.
- **37% in Q3** point to slow or flaky test pipelines — the safety net that has to verify the new volume.



### THE THESIS OF THIS REPORT

AI didn't create most of these problems. It **exposed** them. Teams that already had fast, reliable tests are amplified by AI. Teams with slow, flaky pipelines and unaligned testing strategies experience what Faros calls the acceleration whiplash: more code merged, more bugs to catch, and a review tax that wipes out the speed gain.



## S 08 / WHAT TO DO ABOUT IT

# Five moves that pay for themselves.

Each recommendation is tied to a specific number in this report. They are ordered by leverage — how much of the pain a single move removes.

## 01 Write a one-page testing strategy — and enforce it.

The biggest Spring Boot bottleneck is not flaky tests; it's that **nobody agrees on what to test**. Define which features get unit tests, which get slice tests (`@WebMvcTest`, `@DataJpaTest`), which get `@SpringBootTest` with Testcontainers, and which get end-to-end. Ship the doc. Make it a PR review check.

→ Addresses Q6 #1 (18.95%) + Q6 #4 mock disagreement (15.79%)

## 02 Cut Spring context startups to one per test suite.

Most Spring Boot suites accidentally rebuild the application context dozens of times because slightly different `@MockBean`, `@TestConfiguration`, or property overrides bust the cache. Audit your suite. One context per slice type. Reuse Testcontainers via singleton pattern. *This single change typically halves CI time on a mid-size codebase.*

→ Addresses Q3 build & test times (21.88%) + Q6 context startup (13.68%)

## 03 Make the test pipeline the gate for AI-generated PRs.

If AI generates a 30-file PR, a human cannot review every line. **The test suite must.** That means coverage gates, mutation testing for critical paths, and a contract for AI tooling: it must produce *integration* tests, not just unit tests that mock the world. Stop merging AI code that only adds `@Mock-heavy` unit tests.

→ Addresses Q3 AI-code friction (25.01%) + Q2 AI review burden (10.9%)

## 04 Quarantine flaky tests within 24 hours.

A flaky test trains your team to ignore red builds. Adopt a hard rule: any test that fails intermittently is **tagged @Disabled with a ticket within one working day**, and the owning team has one sprint to fix or delete it. The signal value of the suite recovers almost immediately.

→ Addresses Q3 flaky tests (15.10%) + Q6 flaky integration (16.84%)

## 05 Treat test-data management as a first-class concern.

16.84% of Spring Boot teams point here. The fix is unglamorous and reliable: per-test transaction rollback by default, dedicated builders or fixtures for shared data, and Testcontainers reused across the JVM. Avoid `@DirtiesContext`. Avoid shared mutable state between tests. Avoid “just clean the database” in `@AfterEach`.

→ Addresses Q6 test data (16.84%)

// GUARANTEED DEVELOPER PRODUCTIVITY BOOST

# 50% Faster Spring Boot Builds in 30 Days.

Stop letting AI-generated code turn your team into **exhausted auditors**. Get blazingly fast feedback loops to deploy with **zero fear**.

● 100% RISK-FREE ● 30-DAY GUARANTEE ● MONEY-BACK IF WE MISS

## Our blueprint for 50% faster builds

Five steps. No heavy jargon. Real results black-on-white.

01

### The Unvarnished Report

A ruthless baseline analysis of your project, pipeline, and ways of working to identify hidden productivity killers.

02

### The Build-Time Turbo

Picking low-hanging fruits directly in the code — Spring context caching, parallelization, Testcontainers optimization — for instant flow.

03

### Pipeline Sanitation

Eliminating flaky tests, removing duplicate steps, and adding smart caching to build a rock-solid safety net.

04

### The Strategy Rollout

Handing over a foolproof standard workflow and showing your team the massive time savings black-on-white.

05

### Ultimate Team Enablement

Hands-on mentoring directly in the code to turn your developers into self-sufficient testing experts.



### Want this applied to your pipeline?

Start with Step 1 — The Unvarnished Report.  
[Book a slot →](#)

## Book Step 1 — The Unvarnished Report.

30-minute call. Zero pitch. You leave with at least one concrete optimization you can ship next week — whether we end up working together or not.

Book a slot →

PICK A SLOT

[cal.eu/pragmatech/  
pragmatech-discovery](https://cal.eu/pragmatech/pragmatech-discovery)

## § 10 / ABOUT THIS REPORT

# The whole point of this.

## About PragmaTech

**PragmaTech** helps Java teams build & ship software with confidence — cutting through testing complexity and slow builds for faster feedback and fearless shipping. Founded by **Philip Riecks**, we've spent years inside the engine rooms of startups, scale-ups, and multi-billion-euro enterprises turning slow, flaky test pipelines into fast, trusted safety nets.

This report exists because in 2026, the question changed. It's no longer *"how do we write good tests?"* It's *"how do we validate the 30x more code AI now generates — without our pipeline becoming a maintenance time bomb?"*

## Sources cited

Google Cloud / DORA. *2025 State of AI-Assisted Software Development Report*. 90% AI adoption; AI as "mirror and multiplier"; negative relationship with delivery stability.

Becker, Rush, Barnes & Rein. *Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity*. METR, July 2025. Randomized controlled trial: -19% productivity.

Faros AI. *AI Engineering Report 2026*. Telemetry across 22,000 developers: time-in-PR-review up 441%, incidents-per-PR up 242%.

### NEXT STEP

## Stop the test-pipeline time bomb.

If this report sounds like your week, you are not alone. **Three out of every five engineers surveyed** are losing time to slow test feedback and AI-code friction.

PragmaTech's **Testing Spring Boot Applications** course walks Java teams through the playbook on the previous page — slice tests, Testcontainers, test data, and a shared strategy — in roughly 40 focused lessons. We also run on-site workshops and a 30-day build-speed-up engagement with a 50% guarantee.

[pragmatech.digital](https://pragmatech.digital)



### YOU WILL BE THE FIRST TO KNOW

You were one of 60 engineers who left an email at the end of this survey. The next edition of the report — with year-over-year comparisons and a deeper cut on AI-generated code review patterns — lands in your inbox first.